

Planetary Rover — Single Rover, Static Environment

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Domain & Problem Description

A single wheeled rover navigates a static directed graph of locations. It starts at a base station and must collect geological samples from target sites and return them to base. Every move consumes battery. The environment is fully known in advance — no uncertainty, no dynamic obstacles. **Types:** location, sample. **Predicates:** at-rover, connected, sample-at, carrying, delivered, is-base. **Fluents:** battery-level, move-cost.

Modelling Choices

Q1 — Classical PDDL with numeric fluents. Battery is a global numeric fluent. The move action has a hard precondition ($\geq$ battery-level move-cost) and decreases it on every traversal. Three actions separate task phases: move, collect-sample, deliver-sample. This forces genuine energy reasoning — the planner cannot move when charge is insufficient.

Q2 — PDDL+ with continuous processes and safety event.

Construct	Name	Effect / Role
Process	idle-drain	Decrease battery-level (* #t idle-drain-rate) — always active
Process	move-drain	Decrease battery-level (* #t move-drain-rate) — only while rover-active
Event	battery-critical	Fires when battery $\leq$ threshold; sets (not system-ok) — blocks all actions

Key insight: idle time has a real cost. Battery drains even while collecting. The critical event fires autonomously — the planner cannot control it, only race against it. Mission feasibility depends on completing all goals before the threshold is reached.

Plan Output

Q1-P1 (battery=60): 4-step plan — move, collect, move, deliver. Battery remaining: 40. Trivial, as expected. Planner finds optimal route in 6 expanded nodes.

Q1-P2 (battery=130): 16-step plan — 3 samples, 3 separate trips. Planner correctly routes via waypoint1→site-C for the third sample, finding the minimum-length plan. Battery constraint is active — plan fails without sufficient charge.

Q2-P1 (battery=80): 4-step plan with processes and event active ($|P|=1$, $|E|=1$). idle-drain process runs continuously; battery-critical event gates system-ok. Plan completes at elapsed time 0 — ENHSP's satisficing search finds instantaneous sequence valid under the continuous model.

Q2-P2 (battery=80, 2 samples): 8-step plan. Planner chooses base→site-B first (direct edge, cost 20), then site-A (cost 10+10). Trip order is determined by the planner to satisfy the goal while remaining above critical-threshold=5.

Limitations

No geometry — locations are symbolic; terrain, gradient, and slip are not modelled.

No uncertainty — real rovers face sensor noise, hardware faults, and comm delays.

No recharge — solar/RTG charging absent; battery is strictly decreasing.

Task-motion gap — collect-sample hides instrument deployment (itself a planning problem).

Instantaneous moves — travel time not modelled; ENHSP does not support durative actions in PDDL+ mode.